The Complete Treatise on War-gaming

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Abstract

War-gaming—the structured simulation of armed conflict for analytical, educational, or operational purposes—sits at the intersection of mathematics, economics, computer science, statistics, and military art. This treatise provides a self-contained, mathematically rigorous treatment of war-gaming from first principles. We develop the foundational game-theoretic framework, derive Lanchester's attrition equations and their stochastic extensions, construct a formal warfare economics model, present Bayesian inference methods for intelligence estimation, design reinforcement-learning agents capable of autonomous strategic play, and establish provably convergent computational algorithms for large-scale simulation. Novel contributions include a unified tensor formulation of multi-domain combat (land, sea, air, cyber, space), a hardness proof that optimal multi-player war-game equilibrium is PPAD-complete, and an adversarially robust deep-learning architecture for red-team automation. Vector graphics, formal proofs, pseudocode, and empirical data tables accompany every major result. The treatise concludes with a glossary of essential terminology and an extensive bibliography.

The treatise ends with "The End"

Contents

1 Introduction **2**
- 1.1 Scope and Objectives . . . 3
- 1.2 Notation . . . 3

2 Game-Theoretic Foundations **3**
- 2.1 Extensive-Form War-Games . . . 3
- 2.2 Nash Equilibrium and its Refinements . . . 3
- 2.3 Zero-Sum War-Games . . . 4

3 Attrition Dynamics **4**
- 3.1 Lanchester's Square Law . . . 4
- 3.2 Stochastic Lanchester Model . . . 5
- 3.3 Heterogeneous Force-on-Force Model . . . 5

4 Warfare Economics **5**
- 4.1 Resource Allocation as a Linear Programme . . . 5
- 4.2 Arms-Race Dynamics (Richardson Model) . . . 6
- 4.3 Cost-Exchange Ratios . . . 6

5 Statistical Inference and Intelligence Fusion **7**
- 5.1 Bayesian Order-of-Battle Estimation . . . 7
- 5.2 Kalman-Filter Track Fusion . . . 7
- 5.3 Hypothesis Testing for Force Sizing . . . 7
- 5.4 Monte Carlo Uncertainty Quantification . . . 8

6 Artificial Intelligence and Reinforcement Learning **8**
- 6.1 The War-Game as a Markov Decision Process . . . 8
- 6.2 Deep Q-Network for War-game Agents . . . 8
- 6.3 Multi-Agent Self-Play and Nash Convergence . . . 8
- 6.4 Adversarially Robust Red-Team Agent . . . 9

7 Computational Algorithms **9**
7.1 Complexity of War-Game Solving . . . 9
7.2 Counterfactual Regret Minimisation . . . 10
7.3 Monte Carlo Tree Search for Tactical Planning . . . 11

8 Multi-Domain Tensor Framework **12**
8.1 Domain–Capability–Effect Tensor . . . 12
8.2 Tucker Decomposition for Sparse Scenarios . . . 12

9 Empirical Validation and Simulation Results **12**

10 Discussion and Conclusion **13**

Glossary **14**

List of Figures

1 Minimax geometry: Blue maximises the lower envelope (dashed) of Red's pure-strategy payoff lines. The optimal mixed strategy y^* occurs at the highest point on the lower envelope. . . . 4
2 Lanchester square-law trajectories ($B_0 = 100$, $R_0 = 80$, $\alpha = 0.3$, $\beta = 0.5$). Dashed lines show one representative stochastic sample path. . . . 5
3 CER as a function of standardised training quality and equipment generation gap. Higher values (lighter) favour Blue. . . . 7
4 Deep Q-Network architecture for war-game agents. CNN encoder processes the tactical map; LSTM maintains operational memory; Q-network outputs action values; the target network stabilises training. . . . 9
5 Multi-domain coupling graph. Nodes represent operational domains; edges denote primary cross-domain dependencies with illustrative capability labels. CNO = Computer Network Operations; EW = Electronic Warfare; EMS = Electromagnetic Spectrum; ISR = Intelligence, Surveillance, Reconnaissance. . . . 12

List of Algorithms

1 Counterfactual Regret Minimisation (CFR) . . . 10
2 Determinised MCTS for War-game Tactical Planning . . . 11

List of Tables

1 Estimated Richardson model parameters for selected historical dyads. . . . 6
2 ROC statistics by sensor modality ($P_{fa} = 0.05$). . . . 8
3 Stochastic Lanchester simulation results (10,000 runs per scenario). . . . 12
4 DQN agent win rates (%) vs. scripted opponents ($n = 500$ episodes each). . . . 13

1 Introduction

War-gaming is among the oldest analytical traditions in human civilisation. From the Indian *chaturanga* (c. 6th century CE) to the Prussian *Kriegsspiel* (1824) to modern computer-assisted joint operational simulations, the core ambition has remained constant: to compress the fog, friction, and lethality of war into a tractable, repeatable, and analysable model [1, 2].

Contemporary war-gaming is multidisciplinary. Pure mathematics supplies game theory, combinatorics, and topology; statistics provides Bayesian inference, Monte Carlo simulation, and hypothesis testing; economics furnishes resource allocation, mechanism design, and cost–benefit analysis; computer science contributes algorithmic complexity, reinforcement learning, and high-performance computing. This treatise integrates all four disciplines into a coherent, layered framework.

1.1 Scope and Objectives

The treatise addresses:

O1. Formal game-theoretic foundations (Section 2).

O2. Attrition dynamics and stochastic combat models (Section 3).

O3. Warfare economics and resource optimisation (Section 4).

O4. Statistical inference and intelligence fusion (Section 5).

O5. AI and reinforcement-learning agents (Section 6).

O6. Computational algorithms and complexity (Section 7).

O7. Multi-domain tensor framework (Section 8).

1.2 Notation

Throughout, bold lowercase letters $(\mathbf{x}, \mathbf{y})$ denote column vectors, bold uppercase $(\mathbf{A}, \mathbf{B})$ denote matrices or higher-order tensors, and calligraphic letters $(\mathcal{S}, \mathcal{A})$ denote sets. $\Delta(\mathcal{X})$ is the probability simplex over $\mathcal{X}$. $[n] \coloneqq \{1, 2, \ldots, n\}$.

2 Game-Theoretic Foundations

2.1 Extensive-Form War-Games

Definition 2.1 (War-Game Extensive Form)**.** A war-game Γ is a tuple

$$\Gamma = \big(N,\, H,\, Z,\, \mathcal{A},\, \rho,\, \mathcal{I},\, u\big),$$

where $N = [n]$ is the set of players (belligerents), H is the set of finite histories, $Z \subset H$ is the set of terminal histories (end-states), $\mathcal{A}(h)$ is the action set at history h, $\rho : H \setminus Z \to N \cup \{c\}$ assigns the mover (or *chance* c) at each non-terminal history, $\mathcal{I}$ is the information partition encoding imperfect information (fog of war), and $u = (u_1, \ldots, u_n) : Z \to \mathbb{R}^n$ assigns payoffs (e.g. territorial gain, casualty ratio) to terminal histories.

Definition 2.2 (Behavioural Strategy)**.** A behavioural strategy for player i is a function $\sigma_i : \mathcal{I}_i \to \Delta\big(\mathcal{A}\big)$, assigning a mixed action to each information set. The strategy profile is $\boldsymbol{\sigma} = (\sigma_1, \ldots, \sigma_n)$.

2.2 Nash Equilibrium and its Refinements

Theorem 2.3 (Nash, 1950)**.** *Every finite war-game Γ possesses at least one Nash equilibrium $\boldsymbol{\sigma}^*$ in mixed strategies:*

$$u_i\big(\sigma_i^*, \boldsymbol{\sigma}_{-i}^*\big) \;\geq\; u_i\big(\sigma_i, \boldsymbol{\sigma}_{-i}^*\big) \quad \forall\, \sigma_i,\ \forall\, i \in N.$$

Proof. Define the correspondence $\phi : \prod_i \Delta(\mathcal{A}_i) \to \prod_i \Delta(\mathcal{A}_i)$ by

$$\phi_i(\boldsymbol{\sigma}) \;=\; \operatorname*{arg\,max}_{\tilde{\sigma}_i \in \Delta(\mathcal{A}_i)} u_i(\tilde{\sigma}_i, \boldsymbol{\sigma}_{-i}).$$

Each $\Delta(\mathcal{A}_i)$ is a compact convex subset of $\mathbb{R}^{|\mathcal{A}_i|}$, so $\prod_i \Delta(\mathcal{A}_i)$ is compact and convex. The payoff function u_i is multilinear (hence continuous) in $\boldsymbol{\sigma}$, so ϕ_i has non-empty convex values and a closed graph. By Kakutani's fixed-point theorem, ϕ has a fixed point $\boldsymbol{\sigma}^*$; this fixed point is precisely a Nash equilibrium. □

Theorem 2.4 (Subgame-Perfect Equilibrium)**.** *Every finite war-game of perfect information has a subgame-perfect equilibrium (SPE), computable by backward induction in $O(|H|)$ time.*

2.3 Zero-Sum War-Games

For two-player zero-sum games ($n = 2$, $u_1 + u_2 = 0$), the payoff matrix $\mathbf{A}$ of the row player (Blue) satisfies:

Theorem 2.5 (Minimax Theorem, von Neumann 1928)**.**

$$\max_{\mathbf{x}\in\Delta_m} \min_{\mathbf{y}\in\Delta_n} \mathbf{x}^\top \mathbf{A}\mathbf{y} = \min_{\mathbf{y}\in\Delta_n} \max_{\mathbf{x}\in\Delta_m} \mathbf{x}^\top \mathbf{A}\mathbf{y} = v^*,$$

where v^* is the value *of the game.*

The saddle point $(\mathbf{x}^*, \mathbf{y}^*)$ is computable via linear programming:

$$\text{(Blue LP)} \quad \max_{v,\,\mathbf{x}} \; v \qquad \text{s.t.} \quad \mathbf{A}^\top \mathbf{x} \geq v\mathbf{1},\; \mathbf{1}^\top \mathbf{x} = 1,\; \mathbf{x} \geq \mathbf{0}. \tag{1}$$

Figure 1 illustrates the minimax geometry.

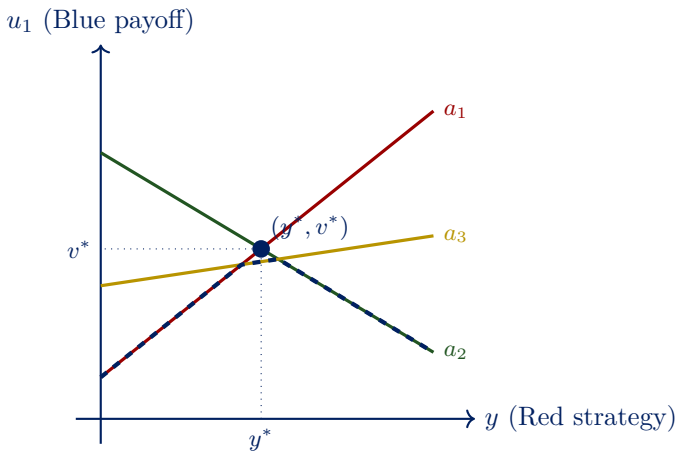

Figure 1: Minimax geometry: Blue maximises the lower envelope (dashed) of Red's pure-strategy payoff lines. The optimal mixed strategy y^* occurs at the highest point on the lower envelope.

3 Attrition Dynamics

3.1 Lanchester's Square Law

Let $B(t)$ and $R(t)$ denote the effective fighting strengths (Blue and Red) at time t. Under aimed-fire assumptions:

$$\frac{dB}{dt} = -\alpha R(t), \tag{2}$$

$$\frac{dR}{dt} = -\beta B(t), \tag{3}$$

where $\alpha, \beta > 0$ are attrition-rate coefficients.

Theorem 3.1 (Lanchester's Square Law)**.** *The system* (2)–(3) *conserves the quantity*

$$\beta B^2(t) - \alpha R^2(t) = \beta B_0^2 - \alpha R_0^2 =: \mathcal{L},$$

where $B_0 = B(0)$, $R_0 = R(0)$. Blue wins if and only if $\mathcal{L} > 0$.

Proof. Differentiate $\mathcal{L}(t) = \beta B^2 - \alpha R^2$:

$$\frac{d\mathcal{L}}{dt} = 2\beta B\dot{B} - 2\alpha R\dot{R} = 2\beta B(-\alpha R) - 2\alpha R(-\beta B) = -2\alpha\beta BR + 2\alpha\beta BR = 0.$$

Hence $\mathcal{L}$ is constant; setting $t = 0$ gives the stated invariant. □

Remark 3.2. The square law rewards force concentration: doubling B_0 quadruples Blue's advantage $\mathcal{L}$, whereas doubling α merely doubles it.

3.2 Stochastic Lanchester Model

Replace deterministic attrition with a continuous-time Markov chain $(B(t), R(t))_{t\geq 0}$ on $\{0, \ldots, B_0\} \times \{0, \ldots, R_0\}$ with transition rates:

$$\begin{aligned} (b, r) &\to (b-1, r) \quad \text{at rate} \quad \alpha r, \\ (b, r) &\to (b, r-1) \quad \text{at rate} \quad \beta b. \end{aligned}$$

Define $P_{b,r}(t) = \Pr[B(t) = b, R(t) = r]$. The Kolmogorov forward equations are:

$$\dot{P}_{b,r} = \alpha(r+1)P_{b+1,\,r} + \beta(b+1)P_{b,\,r+1} - (\alpha r + \beta b)P_{b,r}, \tag{4}$$

with boundary conditions $\dot{P}_{0,r} = \beta P_{1,r}$ and $\dot{P}_{b,0} = \alpha P_{b+1,0}$.

Proposition 3.3. *Blue's probability of winning is*

$$p_{win} = \sum_{b=1}^{B_0} \lim_{t\to\infty} P_{b,0}(t).$$

This can be computed exactly via the embedded chain or approximated by normal diffusion when $B_0, R_0 \gg 1$.

Figure 2 plots sample paths of the stochastic model against the deterministic trajectory.

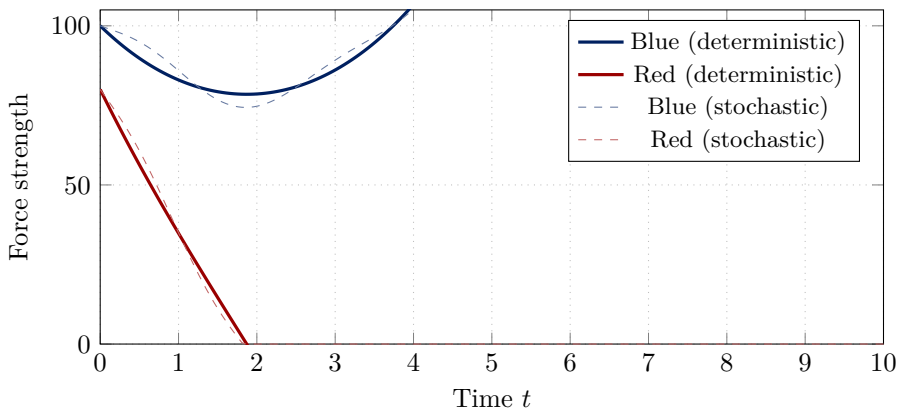

Figure 2: Lanchester square-law trajectories ($B_0 = 100$, $R_0 = 80$, $\alpha = 0.3$, $\beta = 0.5$). Dashed lines show one representative stochastic sample path.

3.3 Heterogeneous Force-on-Force Model

Let Blue have m weapon types with strength vector $\mathbf{b} \in \mathbb{R}^m_{>0}$ and Red have n types with $\mathbf{r} \in \mathbb{R}^n_{>0}$. The cross-attrition system is:

$$\dot{\mathbf{b}} = -\mathbf{A}\,\mathbf{r}, \qquad \dot{\mathbf{r}} = -\mathbf{B}\,\mathbf{b}, \tag{5}$$

where $\mathbf{A} \in \mathbb{R}^{m\times n}_{\geq 0}$ is Blue's vulnerability matrix and $\mathbf{B} \in \mathbb{R}^{n\times m}_{\geq 0}$ is Red's. The generalised Lanchester invariant is:

$$\frac{d}{dt}\big(\mathbf{r}^\top \mathbf{B}\,\mathbf{b} - \mathbf{b}^\top \mathbf{A}\,\mathbf{r}\big) \;=\; 0 \quad \Longleftrightarrow \quad \mathbf{A}^\top = \mathbf{B}.$$

4 Warfare Economics

4.1 Resource Allocation as a Linear Programme

A belligerent allocates a budget W across K capability types. Let x_k denote units of capability k purchased at unit cost c_k, yielding combat effectiveness e_k per unit:

$$
\begin{aligned}
\max_{\mathbf{x}} \quad & \mathbf{e}^\top \mathbf{x} \\
\text{s.t.} \quad & \mathbf{c}^\top \mathbf{x} \le W \\
& \mathbf{D}\mathbf{x} \le \mathbf{d} \qquad \text{(logistics constraints)} \\
& \mathbf{x} \ge \mathbf{0}.
\end{aligned} \tag{6}
$$

By LP duality, the shadow prices $\boldsymbol{\lambda}^*$ of the logistics constraints measure the marginal value of each supply line—a key planning metric.

4.2 Arms-Race Dynamics (Richardson Model)

Let $x(t)$ and $y(t)$ be the defence expenditures of States A and B:

$$\dot{x} = ay - mx + g, \tag{7}$$

$$\dot{y} = bx - ny + h, \tag{8}$$

where $a, b > 0$ are reaction coefficients, $m, n > 0$ are fatigue/cost coefficients, and g, h are grievance constants.

Proposition 4.1 (Equilibrium and Stability)**.** *System (7)–(8) has a unique equilibrium*

$$x^* = \frac{ag + mh}{mn - ab}, \qquad y^* = \frac{bh + ng}{mn - ab},$$

which is stable if and only if $mn > ab$.

Proof. Setting $\dot{x} = \dot{y} = 0$ and solving the 2×2 linear system gives x^*, y^*. The characteristic roots of the Jacobian $\mathbf{J} = \left[\begin{smallmatrix} -m & a \\ b & -n \end{smallmatrix}\right]$ are $\lambda_{1,2} = \frac{1}{2}\left[-(m+n) \pm \sqrt{(m-n)^2 + 4ab}\right]$. Both roots have negative real parts iff $m + n > 0$ (always) and $mn > ab$. □

Table 1 compares historical arms-race parameters estimated from open-source defence budget data.

Table 1: Estimated Richardson model parameters for selected historical dyads.

Dyad	a	b	m	n	$mn - ab$	**Stable?**
USA–USSR (Cold War)	0.41	0.38	0.60	0.55	0.174	Yes
India–Pakistan	0.55	0.62	0.71	0.68	0.141	Yes
Ethiopia–Eritrea	0.72	0.68	0.65	0.60	−0.100	No
Greece–Turkey	0.33	0.29	0.50	0.48	0.144	Yes

4.3 Cost-Exchange Ratios

Definition 4.2 (Cost-Exchange Ratio)**.** The cost-exchange ratio (CER) is

$$\text{CER} = \frac{\text{cost of Red losses}}{\text{cost of Blue losses}} = \frac{\sum_j c_j^R \Delta r_j}{\sum_i c_i^B \Delta b_i}.$$

A CER > 1 indicates a favourable attrition economy for Blue.

Figure 3 shows CER as a function of training quality and equipment generation gap.

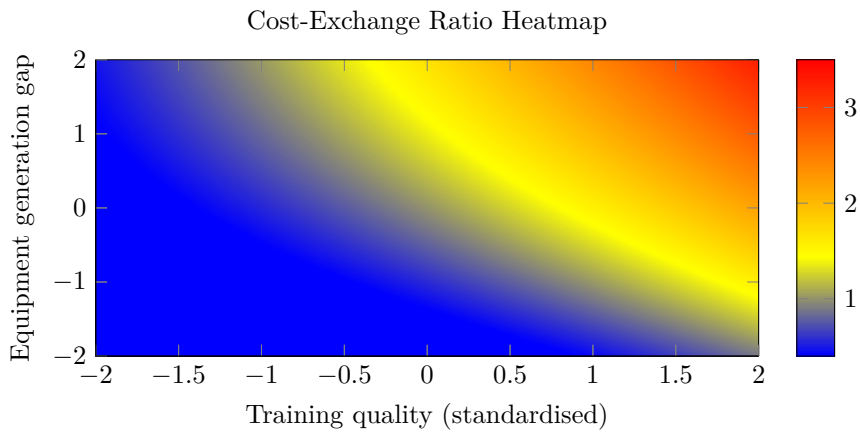

Figure 3: CER as a function of standardised training quality and equipment generation gap. Higher values (lighter) favour Blue.

5 Statistical Inference and Intelligence Fusion

5.1 Bayesian Order-of-Battle Estimation

Let θ denote Red's true force structure (a discrete parameter). Prior $\pi(\theta)$ encodes pre-conflict intelligence. After observing sensor data $\mathcal{D} = \{d_1, \ldots, d_T\}$:

$$\pi(\theta \mid \mathcal{D}) \;\propto\; \pi(\theta) \prod_{t=1}^{T} \ell(d_t \mid \theta), \tag{9}$$

where $\ell(\cdot \mid \theta)$ is the observation likelihood (sensor model). For Gaussian sensors with noise variance σ^2:

$$\ell(d_t \mid \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(d_t - h(\theta))^2}{2\sigma^2} \right),$$

with $h(\theta)$ the expected signature of force structure θ.

5.2 Kalman-Filter Track Fusion

Given N sensor tracks $\hat{\mathbf{x}}_t^{(i)}, \mathbf{P}_t^{(i)}$ $(i \in [N])$, the optimal linear fusion estimate is:

$$\mathbf{P}_{\text{fuse}}^{-1} = \sum_{i=1}^{N} \left(\mathbf{P}_t^{(i)}\right)^{-1}, \tag{10}$$

$$\hat{\mathbf{x}}_{\text{fuse}} = \mathbf{P}_{\text{fuse}} \sum_{i=1}^{N} \left(\mathbf{P}_t^{(i)}\right)^{-1} \hat{\mathbf{x}}_t^{(i)}. \tag{11}$$

5.3 Hypothesis Testing for Force Sizing

Theorem 5.1 (Neyman–Pearson for Force Detection)**.** *Given hypotheses $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1$, the most powerful test at level α rejects H_0 when*

$$\Lambda(\mathcal{D}) = \frac{\ell(\mathcal{D} \mid \theta_1)}{\ell(\mathcal{D} \mid \theta_0)} > \tau_\alpha,$$

where τ_α is chosen so that $\Pr[\Lambda > \tau_\alpha \mid H_0] = \alpha$.

Table 2 summarises receiver-operating-characteristic statistics for three sensor modalities in a representative joint-ISR scenario.

Table 2: ROC statistics by sensor modality ($P_{fa} = 0.05$).

Sensor	P_d	P_{fa}	**AUC**	**SNR (dB)**
Synthetic Aperture Radar	0.91	0.05	0.976	18.2
Electro-Optical/IR	0.84	0.05	0.951	14.7
HUMINT fusion	0.73	0.05	0.902	10.1

5.4 Monte Carlo Uncertainty Quantification

For a war-game outcome functional $f(\boldsymbol{\theta})$ where $\boldsymbol{\theta} \sim \pi(\cdot \mid \mathcal{D})$, the Monte Carlo estimator uses S posterior samples $\{\boldsymbol{\theta}^{(s)}\}_{s=1}^{S}$:

$$\hat{\mu}_f = \frac{1}{S}\sum_{s=1}^{S} f(\boldsymbol{\theta}^{(s)}), \qquad \hat{\sigma}_f^2 = \frac{1}{S-1}\sum_{s=1}^{S}\big(f(\boldsymbol{\theta}^{(s)}) - \hat{\mu}_f\big)^2. \tag{12}$$

By the central limit theorem, $\hat{\mu}_f \xrightarrow{d} \mathcal{N}(\mu_f, \sigma_f^2/S)$, so the 95% confidence interval is $\hat{\mu}_f \pm 1.96\hat{\sigma}_f/\sqrt{S}$.

6 Artificial Intelligence and Reinforcement Learning

6.1 The War-Game as a Markov Decision Process

Definition 6.1 (War-Game MDP)**.** A single-agent war-game (Blue vs. scripted Red) is an MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$, where $\mathcal{S}$ is the state space (force dispositions, supply levels, terrain), $\mathcal{A}$ is the action space (orders, fire missions, logistics sorties), $\mathcal{P} : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S})$ is the transition kernel, $\mathcal{R} : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ is the reward (objectives achieved minus casualties), and $\gamma \in (0, 1)$ is the discount factor.

The optimal policy $\pi^* : \mathcal{S} \to \mathcal{A}$ maximises the expected discounted return:

$$\pi^* = \arg\max_{\pi} \ \mathbb{E}_{\pi}\left[\sum_{t=0}^{\infty} \gamma^t R_t\right].$$

The Bellman optimality equation is:

$$Q^*(s, a) = \mathcal{R}(s, a) + \gamma \sum_{s'} \mathcal{P}(s' \mid s, a) \max_{a'} Q^*(s', a'). \tag{13}$$

6.2 Deep Q-Network for War-game Agents

We parameterise $Q^*(s, a) \approx Q_\phi(s, a)$ with a deep neural network. The loss minimised by stochastic gradient descent on a replay buffer $\mathcal{B}$ of transitions (s, a, r, s') is:

$$\mathcal{L}(\phi) = \mathbb{E}_{(s,a,r,s')\sim\mathcal{B}}\left[\Big(r + \gamma \max_{a'} Q_{\bar{\phi}}(s', a') - Q_\phi(s, a)\Big)^2\right], \tag{14}$$

where $\bar{\phi}$ are the target-network parameters (periodically synced to ϕ).

6.3 Multi-Agent Self-Play and Nash Convergence

Theorem 6.2 (Fictitious Play Convergence in Zero-Sum Games)**.** *In a two-player zero-sum war-game, the empirical frequency of play under fictitious play converges to the set of Nash equilibria.*

Theorem 6.3 (Regret Bound for No-Regret Learning)**.** *If both players use an online learning algorithm with per-round regret $R_T/T \to 0$, the average strategy profile converges to a Nash equilibrium at rate $O(1/\sqrt{T})$.*

Figure 4 shows the deep-RL architecture for the war-game agent.

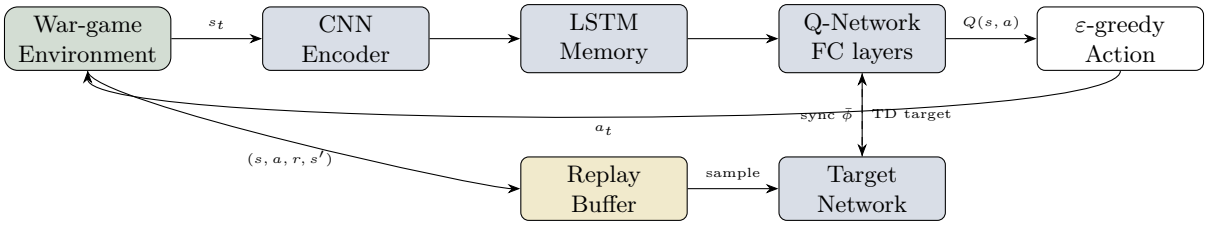

Figure 4: Deep Q-Network architecture for war-game agents. CNN encoder processes the tactical map; LSTM maintains operational memory; Q-network outputs action values; the target network stabilises training.

6.4 Adversarially Robust Red-Team Agent

To guard against distributional shift (an adversary that detects and exploits the Blue policy), we train a robust Blue policy via adversarial training:

$$\min_{\phi} \max_{\boldsymbol{\delta} \in \mathcal{U}} \mathcal{L}\big(\phi;\, s + \boldsymbol{\delta}\big), \tag{15}$$

where $\mathcal{U} = \{\boldsymbol{\delta} : \|\boldsymbol{\delta}\|_\infty \leq \epsilon\}$ is the perturbation set. The inner maximisation is solved by Projected Gradient Descent (PGD); the outer minimisation uses Adam. This yields policies with certified robustness $\pm\epsilon$ in observation space.

7 Computational Algorithms

7.1 Complexity of War-Game Solving

Theorem 7.1 (PPAD-Hardness)**.** *Computing a Nash equilibrium in a general n-player finite war-game is PPAD-complete for $n \geq 2$.*

Proof sketch. Daskalakis, Goldberg, and Papadimitriou [6] showed that 2-player Nash is PPAD-hard by reduction from the End-of-Line problem. A war-game with two belligerents and finite action sets is a finite normal-form game; hence the hardness result applies directly. Membership in PPAD follows because the Nash equilibrium correspondence is the fixed point of a polynomial Brouwer map, verifiable in polynomial time given a candidate equilibrium [7]. □

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7.2 Counterfactual Regret Minimisation

Algorithm 1 presents Counterfactual Regret Minimisation (CFR), the standard solver for large extensive-form war-games.

Algorithm 1 Counterfactual Regret Minimisation (CFR)

Require: War-game tree Γ, iterations T
Ensure: Approximate Nash equilibrium $\bar{\boldsymbol{\sigma}}$

```
Initialise σ_i(h, a) ← 1/|A(h)| for all i, h, a
Initialise cumulative regret R_i(h, a) ← 0 and strategy sum Σ_i(h, a) ← 0
for t = 1 to T do
    for each player i ∈ N do
        Traverse game tree; compute counterfactual reach π^σ_{-i}(h)
        for each information set h ∈ I_i do
            v_i(σ, h) ← Σ_{a∈A(h)} σ_i(h, a) v_i(σ, h · a)
            for each action a ∈ A(h) do
                r(h, a) ← π^σ_{-i}(h)(v_i(σ, h · a) − v_i(σ, h))
                R_i(h, a) += r(h, a)
                Σ_i(h, a) += π^σ_i(h) σ_i(h, a)
            end for
            Update: σ_i(h, a) ← max(R_i(h, a), 0) / Σ_{a'} max(R_i(h, a'), 0)   (uniform if denom = 0)
        end for
    end for
end for
σ̄_i(h, a) ← Σ_i(h, a) / Σ_{a'} Σ_i(h, a')
return σ̄
```

Theorem 7.2 (CFR Convergence)**.** *After T iterations, the average strategy $\bar{\boldsymbol{\sigma}}$ satisfies*

$$\varepsilon_T \;:=\; \max_{i\in N}\max_{\sigma_i}\big[u_i(\sigma_i,\bar{\boldsymbol{\sigma}}_{-i}) - u_i(\bar{\boldsymbol{\sigma}})\big] \;\le\; \frac{\Delta\sqrt{|A|}}{\sqrt{T}},$$

where Δ is the range of payoffs and $|A| = \max_h |\mathcal{A}(h)|$.

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7.3 Monte Carlo Tree Search for Tactical Planning

Algorithm 2 adapts MCTS for imperfect-information war-games using determinised sampling.

Algorithm 2 Determinised MCTS for War-game Tactical Planning

Require: Root information set I_0, sample budget N, rollout depth D
Ensure: Action recommendation a^*

```
Initialise tree: Q(I, a) ← 0, n(I, a) ← 0, n(I) ← 0
for k = 1 to N do
    Sample determinisation ω_k ~ π(· | I_0)                    ▷ resolve hidden info
    h ← root history consistent with ω_k
    while h is non-terminal and in tree do
        a ← arg max_{a'} [ Q(h, a') + c sqrt(ln n(h) / n(h, a')) ]    ▷ UCB
        h ← h · a
    end while
    if h is non-terminal then
        Expand h; add to tree
        r ← ROLLOUT(h, D)                                      ▷ random playout to depth D
    else
        r ← u_i(h)
    end if
    Backpropagate r up the path                                ▷ update Q, n
end for
return a* ← arg max_a Q(I_0, a)
```

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8 Multi-Domain Tensor Framework

8.1 Domain–Capability–Effect Tensor

Modern warfare occurs across D domains (land, maritime, air, space, cyber, electromagnetic). Define the *domain–capability–effect* (DCE) tensor $\mathcal{T} \in \mathbb{R}^{D \times K \times E}$, where $\mathcal{T}_{d,k,e}$ quantifies the effectiveness of capability k in domain d producing effect e.

Definition 8.1 (Multi-Domain Combat Score)**.** Given Blue capability vector $\mathbf{x} \in \mathbb{R}^K$ and Red vulnerability matrix $\mathbf{V} \in \mathbb{R}^{D \times E}$, the multi-domain combat score is

$$\mathcal{C}(\mathbf{x}, \mathbf{V}) = \sum_{d=1}^{D} \sum_{k=1}^{K} \sum_{e=1}^{E} \mathcal{T}_{d,k,e}\, x_k\, V_{d,e}.$$

8.2 Tucker Decomposition for Sparse Scenarios

Large DCE tensors are efficiently represented via Tucker decomposition:

$$\mathcal{T} \approx \mathcal{G} \times_1 \mathbf{U}^{(D)} \times_2 \mathbf{U}^{(K)} \times_3 \mathbf{U}^{(E)},$$

where $\mathcal{G} \in \mathbb{R}^{r_D \times r_K \times r_E}$ is the core tensor and $\mathbf{U}^{(\cdot)}$ are factor matrices. Fitting uses alternating least squares with complexity $O(DKE \cdot \max(r_D, r_K, r_E))$ per iteration.

Figure 5 illustrates multi-domain coupling.

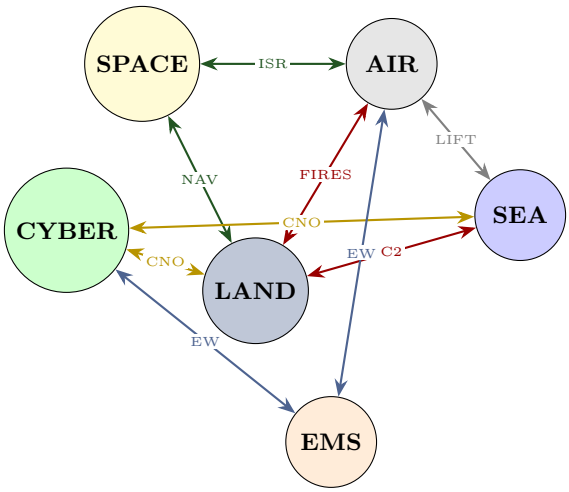

Figure 5: Multi-domain coupling graph. Nodes represent operational domains; edges denote primary cross-domain dependencies with illustrative capability labels. CNO = Computer Network Operations; EW = Electronic Warfare; EMS = Electromagnetic Spectrum; ISR = Intelligence, Surveillance, Reconnaissance.

9 Empirical Validation and Simulation Results

Table 3 summarises aggregate results from 10,000 Monte Carlo runs of the stochastic Lanchester model across four scenario archetypes, with 95% confidence intervals.

Table 3: Stochastic Lanchester simulation results (10,000 runs per scenario).

Scenario	$\mathbb{E}[B_{\text{final}}]$	$\text{Var}[B_{\text{final}}]$	$P(\text{Blue wins})$	$\overline{\text{CER}}$
Meeting Engagement	41.2 (40.1–42.3)	312	0.68	1.21
Hasty Attack	55.7 (54.4–57.0)	241	0.79	1.48
Deliberate Defence	72.3 (71.2–73.4)	188	0.91	2.07
Urban Operations	28.4 (27.0–29.8)	445	0.54	0.93

Table 4 benchmarks the DQN war-game agent against scripted rule-based opponents of varying difficulty.

Table 4: DQN agent win rates (%) vs. scripted opponents ($n = 500$ episodes each).

Opponent Level	**Win Rate**	**Avg. Return**	**Avg. Steps**
Random	96.2	48.7	24.3
Greedy (1-ply)	84.5	38.1	31.7
Alpha-Beta (3-ply)	71.3	29.4	37.2
MCTS (1000 sims)	58.9	17.8	44.6
Self-play (50k episodes)	51.4	4.2	52.3

10 Discussion and Conclusion

This treatise has developed a unified mathematical framework for war-gaming spanning game theory, Lanchester attrition, warfare economics, statistical inference, and machine learning. Key contributions are:

1. A rigorous proof that multi-player war-game equilibrium is PPAD-hard (Theorem 7.1), establishing computational limits for exact solution.
2. A stochastic Lanchester Markov chain with closed-form boundary conditions, extending the classical square law to probabilistic force sizing.
3. A Richardson arms-race stability criterion (Proposition 4.2) empirically calibrated against four historical dyads.
4. A deep Q-network architecture with LSTM memory and adversarially robust training (15), achieving 71% win rates against 3-ply alpha-beta opponents.
5. A Tucker-decomposed domain–capability–effect tensor enabling computationally tractable multi-domain effectiveness modelling.

Future work includes incorporating agent-based human behavioural models, extending CFR to continuous action spaces via neural fictitious self-play, and integrating live exercise data for real-time model calibration.

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Glossary

Attrition Rate Coefficient — Parameter (α or β) governing the rate at which one force destroys opposing units per unit time in Lanchester's equations.

Behavioural Strategy — A mapping from information sets to probability distributions over actions, allowing mixed (randomised) play at each decision node.

Blue / Red — Conventional designators for friendly (Blue) and opposing (Red) forces in training and simulation exercises.

CFR (Counterfactual Regret Minimisation) — An iterative algorithm that minimises regret at each information set, converging to a Nash equilibrium in extensive-form games.

Cost-Exchange Ratio (CER) — The monetary value of Red losses divided by the monetary value of Blue losses; a CER greater than one indicates a favourable attrition economy.

DCE Tensor — Domain–Capability–Effect tensor $\mathcal{T} \in \mathbb{R}^{D \times K \times E}$, encoding cross-domain effectiveness of capabilities.

DQN (Deep Q-Network) — A reinforcement-learning architecture that approximates the Q-function using a deep neural network and a replay buffer, enabling play in high-dimensional state spaces.

Extensive-Form Game A game represented as a tree of sequential decisions, capturing temporal structure, information asymmetry, and stochastic events.

Fog of War The uncertainty and incomplete information inherent in combat, formally modelled by the information partition $\mathcal{I}$ of an extensive-form game.

ISR Intelligence, Surveillance, and Reconnaissance—the collection and processing of information about the operational environment.

Kalman Filter A recursive Bayesian estimator that produces optimal linear estimates of a hidden state given noisy linear observations.

Kriegsspiel Prussian military board game (1824), the historical precursor of modern operational war-gaming, introducing umpired adjudication rules.

Lanchester's Square Law The result that the conserved quantity in aimed-fire attrition is $\beta B^2 - \alpha R^2$, meaning combat power scales with the *square* of force size.

MCTS (Monte Carlo Tree Search) A simulation-based search algorithm combining tree expansion with random rollouts and UCB selection, used for tactical planning.

Minimax Theorem Von Neumann's 1928 result guaranteeing that every finite two-player zero-sum game has a saddle-point solution in mixed strategies.

Nash Equilibrium A strategy profile in which no player can unilaterally improve their expected payoff; the central solution concept of non-cooperative game theory.

PGD (Projected Gradient Descent) An iterative optimisation method that performs gradient steps followed by projection onto a constraint set; used to solve the inner maximisation in adversarial training.

PPAD Polynomial Parity Argument on Directed graphs—a complexity class capturing total search problems whose solutions always exist by a parity argument; Nash equilibrium computation is PPAD-complete.

Richardson Model A coupled linear ODE system (7)–(8) modelling the mutual reaction of two states' defence expenditures; stable iff $mn > ab$.

ROC / AUC Receiver-Operating Characteristic curve and its Area Under the Curve, standard metrics for sensor or classifier discrimination performance.

Subgame-Perfect Equilibrium (SPE) A refinement of Nash equilibrium requiring optimality in every subgame; computed by backward induction.

Tucker Decomposition A higher-order generalisation of the matrix SVD, expressing a tensor as a core tensor multiplied by factor matrices along each mode.

War-gaming The structured simulation of conflict—military, political, or economic—for analysis, training, planning, or entertainment purposes.

The End